6) Presentation Layer:-

→ Encryption, compression, Encoding, Data Translation, SSL, ASCII

→ Converts data into formats (JPEG, MP3, MP4, ASCII, Unicode)

→ Handles encryption & compression (SSL, TLS)

Ensures datas are in a usable format and data encryption occurs.

7)

Application Layer:

→ HTTP, FTP, DNS, SMTP, Browser, User interface, APIs direct interaction with the user and applications support web browsing, email, file transfer

→ Human interaction layer where applications can access the network services.

Networking

→ Universal language for computer networking.

① OSI (7 Layers)

↳ open systems Interconnection

↓

Model is a conceptual framework developed by ISO (International organization for standardization)

1) physical layer:

→ Bits, Transmissions, cables, Hubs, Modulation, Encoding, signal, voltage converts data into electrical radio or optical signals.

→ Manages cables, hubs, repeaters.

→ Transmits raw bit stream over the Physical medium.

→ Wifi-signals, signal interference, weak signal strength (or) faulty antennas.

4) Transport Layer: → reordering packets, handles sequencing → use DHCP

→ Segmentation, TCP, UDP, port No, Flow control, Reliability. ~~ensures~~

→ Ensures end to end communication.

→ Uses TCP (reliable) and UDP (fast, connection less)

→ Transmits Data using transmission protocols including TCP & UDP.

x^n → 2, 6

5) Session Layer:

→ Session Management, Authentication, Synchronization, checkpoints Manage sessions (start, maintain & end)

→ Handles authetication and synchronization

→ Maintains connections & responsible for controlling ports and sessions.

6)